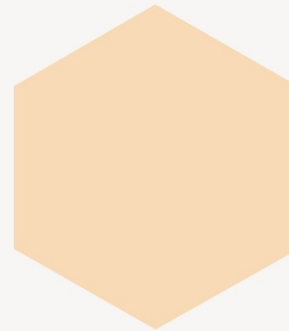
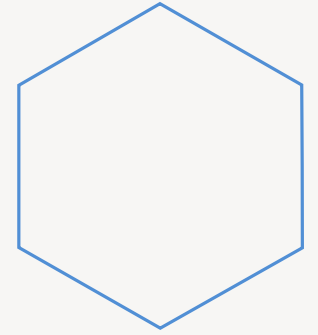


IDS 410: Term Project

Wendy
Alpizar



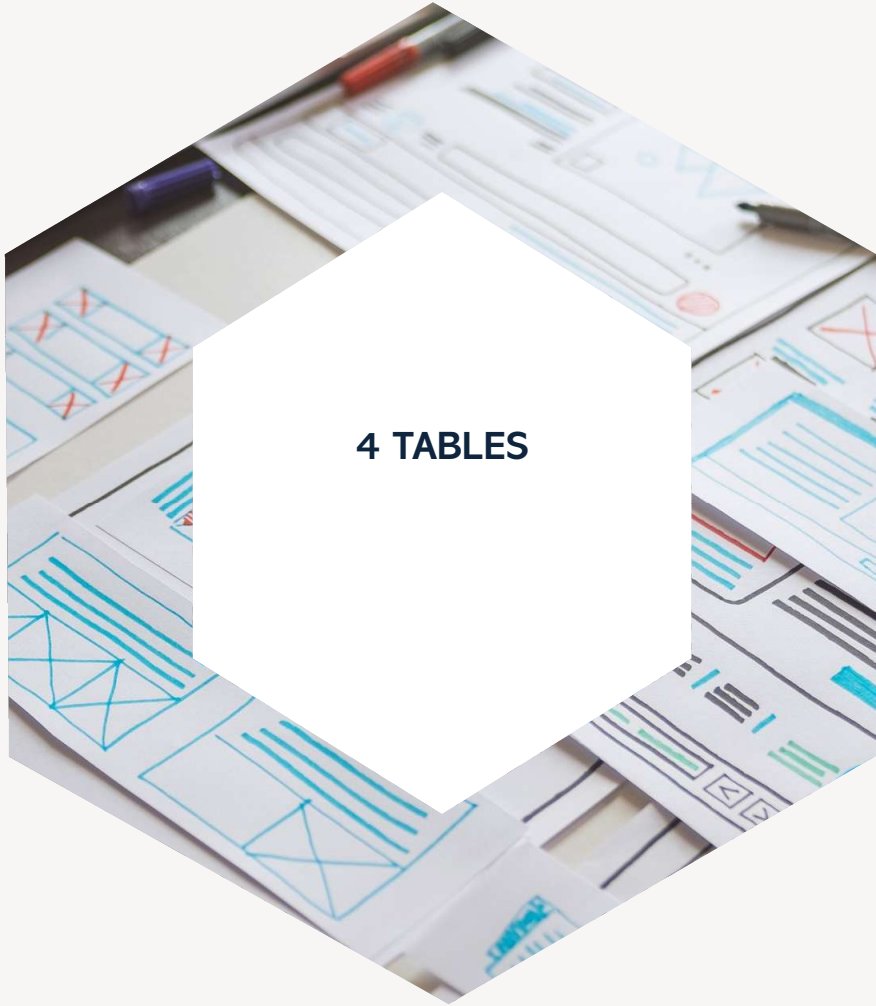
Business Description

My business is a small dog grooming salon called Chihuawow grooming. I will provide many services to make dogs beautiful. The salons main services will include basic baths, full grooming, nail trims, de-shedding treatments and add-ons like teeth cleaning and flea baths. Clients will be able to book appointments for their pets and drop them off at the salon and pick them up when the service is complete. Pricing will vary depending on breed, size, coat type and service package.

From a business standpoint as an owner it's important I track the pets, owner and appointments. Every customer can have one or more dogs. Every appointment is scheduled for a specific dog and a grooming service, the date and time of appointment, along with the groomer who performed the service and the total charge. I also will record the appointment status (Scheduled, Completed, Canceled, No-SHOW) and the check-in/check-out times.

The business wants to use the data for day-to-day operations but also for decision making. The database designed will ensure all business operations run smoothly while allowing us to create meaningful business queries. Chihuawow grooming can gain meaningful insight in customer behavior and services demanded.

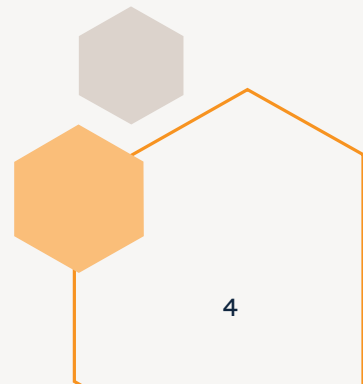




Metadata Tables

Customer_T

Attribute	Data Type	PK/FK	Null?	Description
CustomerID	AUTONUMBER	PK	No	Unique ID for each customer
FirstName	TEXT(20)	—	No	Customer first name
LastName	TEXT(20)	—	No	Customer last name
Phone	TEXT(15)	—	Yes	Phone number
Email	TEXT(50)	—	Yes	Email address

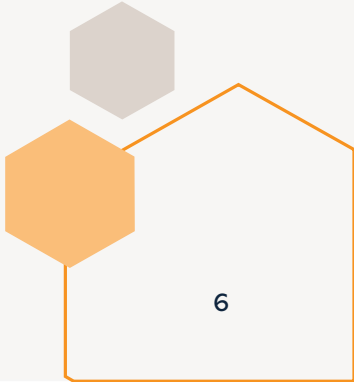


Pet_T

Attribute	Data Type	PK/FK	Null?	Description
PetID	AUTONUMBER	PK	No	Unique ID for each pet
CustomerID	NUMBER	FK	No	Links pet to owner (Customer_T)
PetName	TEXT(20)	—	No	Name of the pet
Breed	TEXT(30)	—	Yes	Dog breed
Weight	NUMBER	—	Yes	Pet's weight (lbs)
CoatType	TEXT(20)	—	Yes	Coat description
BehaviorNotes	TEXT(100)	—	Yes	Temperament, allergies, special instructions

Service_T

Attribute	Data Type	PK/FK	Null?	Description
ServiceID	AUTONUMBER	PK	No	Unique ID for each service
ServiceName	TEXT(30)	—	No	Name of service
ServiceCategory	TEXT(20)	—	Yes	Category (grooming, bath, add-on)
BasePrice	CURRENCY	—	No	Standard price for service
DurationMin	NUMBER	—	Yes	Typical duration in minutes



Appointment_T

Attribute	Data Type	PK/FK	Null?	Description
AppointmentID	AUTONUMBER	PK	No	Unique ID for appointment
PetID	NUMBER	FK	No	Links appointment to a pet
ServiceID	NUMBER	FK	No	Links appointment to a service
AppointmentDateTi me	DATE/TIME	—	No	Date and time of appointment
Status	TEXT(10)	—	No	Scheduled, Completed, Canceled, No-Show
CheckInTime	DATE/TIME	—	Yes	Actual check-in time
CheckOutTime	DATE/TIME	—	Yes	Actual check-out time
TotalCharge	CURRENCY	—	No	Final total amount charged

Relation Schema

Customer_T (CustomerID, FirstName, LastName, Phone, Email)

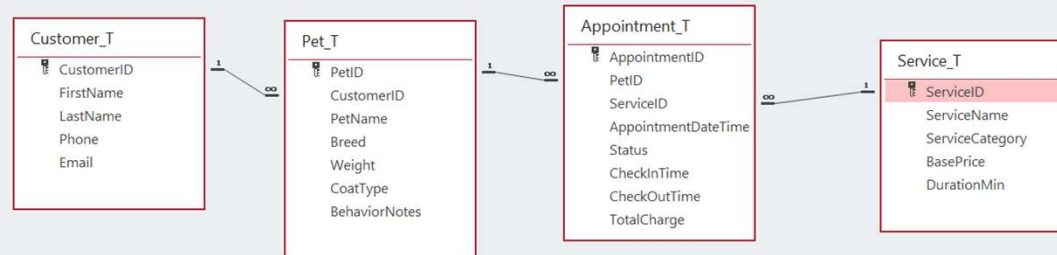
Pet_T (PetID, CustomerID, PetName, Breed, Weight, CoatType, Behavior notes)

Service_T (ServiceID, ServiceName, ServiceCategory, BasePrice, DurationMin)

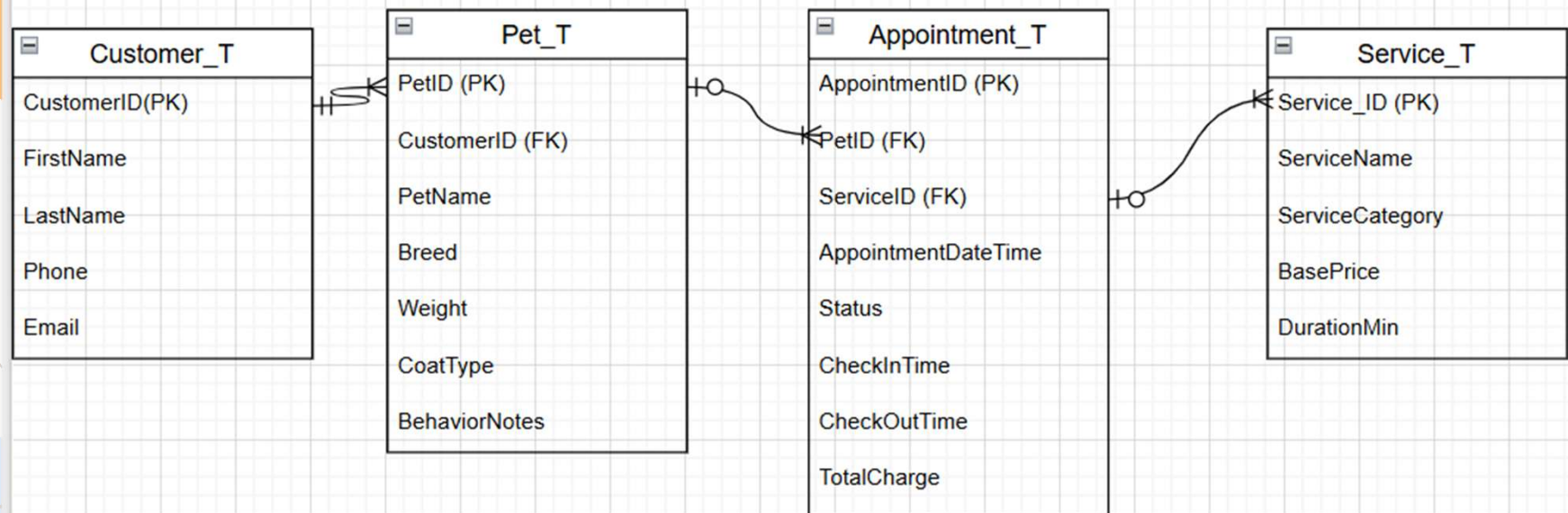
Appointment_T (AppointmentID, PetID, ServiceID, AppointmentDateTime, Status, CheckInTime, CheckOutTime, TotalCharge)

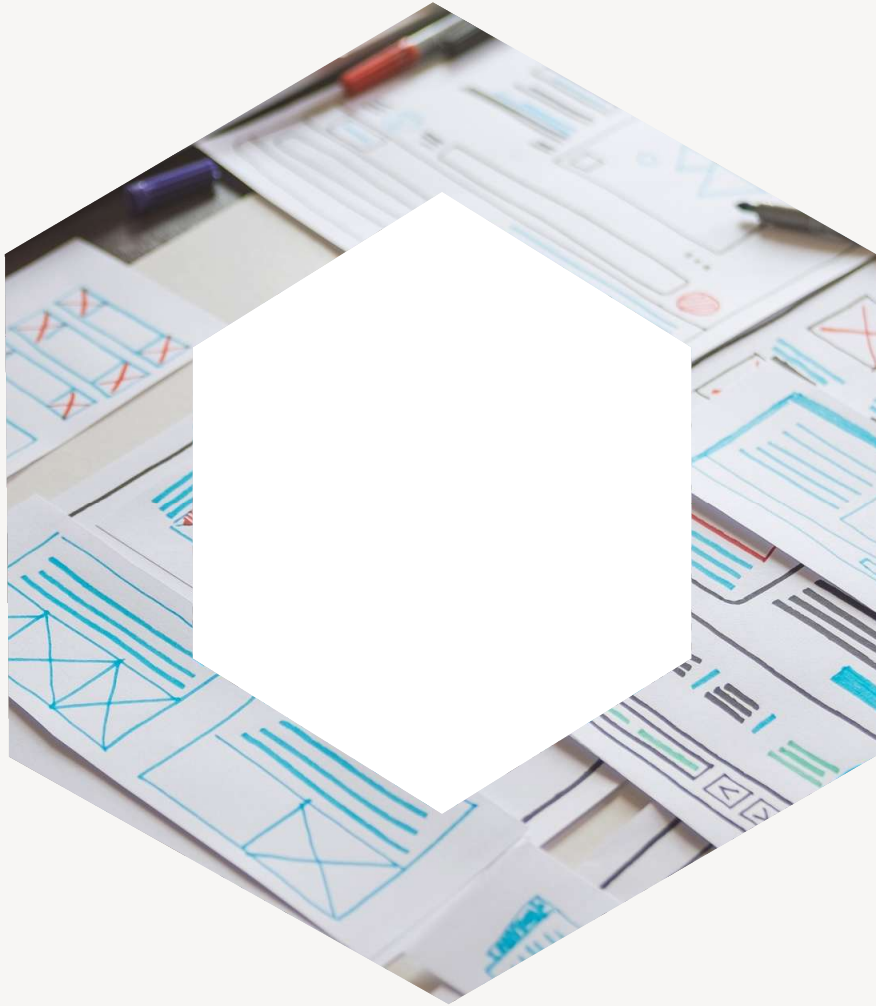
PK: Appointment
FK1: PetID, ServiceID

Database Relationships



EERD





Screenshots of Data and Tables

Customer Table

Customer_T X						
Pet_T X						
Service_T X						
Appointment_T X						
	CustomerID	FirstName	LastName	Phone	Email	Click to Add
+	1	Maria	Lopez	333-333-2222	maria.lopez@gmail.co	
+	2	Dan	Ruiz	223-234-5555	druiz24@yahoo.com	
+	3	Sophia	Kim	630-344-2345	sophia.kim@gmail.co	
+	4	Jason	Patel	234-235-7854	jpatel89@hotmail.cor	
+	5	Emily	Garcia	499-223-2343	emily.garcia@gmail.c	
+	6	Reggie	Kaverga	233-090-7492	rsmith@gmail.com	
+	7	Ana	Torres	220-000-2391	ana.torres@yahoo.cc	
+	8	Kevin	Johnson	815-283-0338	kevin.johnson@gmail	
+	9	Olivia	Martinez	290-023-2239	olivia.mtz@gmail.com	
+	10	Araceli	Garcia	224-265-7751	wendy.3000@hotmai	
*	(New)					

Pet Table

	PetID	CustomerID	PetName	Breed	Weight	CoatType	BehaviorNotes
+	1	1	Chiquita	Chihuahua	8	Short	Nervous, and anxious
+	2	1	Max	Pomeranian	12	Long	Barks
+	3	2	Rocky	Pit Bull Mix	55	Short	
+	4	3	Bella	Golden Retriever	65	Double	Loves water
+	5	4	Frijol	Shih Tzu	14	Long	Squirmy
+	6	5	Coco	Mini Poodle	16	Curly	Friendly
+	7	6	Nala	Mixed	18	Medium	Howls and talkative
+	8	7	Duke	German Shepherd	80	Double	Protective
+	9	8	Milo	Beagle	25	Short	Friendly, howls a lot
+	10	9	Lili	Husky	52	Double	Heavy Shedder
+	11	9	Luna	Weiner Dog	9	Short	Grumpy cute old lady
+	12	10	Lulu	Bichon Frise	10	Curly	Best cutest dog
+	13	10	Athena	Chihuahua	12	Double	Hates Squirrels
*	(New)						

Service Table

	ServiceID ▼	ServiceName ▼	ServiceCateg ▼	BasePrice ▼	DurationMin ▼
+	1	Basic Bath	Bath	\$35.00	45
+	2	Full Groom	Groom	\$65.00	90
+	3	Nail Trim	Add-On	\$15.00	15
+	4	Teeth Cleaning	Add-On	\$20.00	20
+	5	Flea Bath	Treatment	\$40.00	60

Appointment Table

AppointmentID	PetID	ServiceID	AppointmentDateTime	Status	CheckInTime	CheckOutTime	TotalCharged
1	1	2	11/10/2024 10:00:00 AM	Completed	11/10/2024 9:55:00 AM	11/10/2024 11:15:00 AM	\$70.00
2	3	1	11/11/2024 2:00:00 PM	Completed	11/11/2024 1:50:00 PM	11/11/2024 2:50:00 PM	\$35.00
3	5	4	11/12/2024 9:00:00 AM	Completed	11/12/2024 8:55:00 AM	11/12/2024 9:50:00 AM	\$30.00
4	6	2	11/12/2024 11:30:00 AM	Completed	11/12/2024 11:20:00 AM	11/12/2024 1:00:00 PM	\$65.00
5	7	3	11/13/2024 3:00:00 PM	Completed	11/13/2024 2:55:00 PM	11/13/2024 3:20:00 PM	\$15.00
6	10	2	11/14/2024 10:30:00 AM	Completed	11/14/2024 10:20:00 AM	11/14/2024 12:00:00 PM	\$75.00
7	9	1	11/14/2024 1:00:00 PM	Completed	11/14/2024 12:55:00 PM	11/14/2024 1:45:00 PM	\$35.00
8	2	1	11/15/2024 9:00:00 AM	No-Show			\$0.00
9	8	5	11/15/2024 10:30:00 AM	Completed	11/15/2024 10:25:00 AM	11/15/2024 10:55:00 AM	\$20.00
10	1	3	11/16/2024 11:00:00 AM	Completed	11/16/2024 10:55:00 AM	11/16/2024 11:20:00 AM	\$15.00
11	11	1	11/16/2024 11:00:00 AM	Completed	11/17/2024 10:55:00 AM	11/17/2024 11:20:00 AM	\$20.00
12	6	5	11/18/2024 9:30:00 AM	Completed	11/18/2024 9:25:00 AM	11/18/2024 9:25:00 AM	\$20.00
13	12	2	11/19/2024 1:30:00 PM	Completed	11/19/2024 1:25:00 PM	11/19/2024 2:20:00 PM	\$65.00
14	13	2	11/19/2024 1:25:00 PM	Completed	11/19/2024 1:25:00 PM	11/19/2024 2:20:00 PM	\$65.00



Queries

Query #1

OUTPUT

CustomerID	FirstName	LastName	TotalSpent
10	Araceli	Garcia	\$130.00



Problem: Figure out what customer spent the most and by how much.

```
SELECT
  TOP 1 C.CustomerID,
  C.FirstName,
  C.LastName,
  SUM(A.TotalCharge) AS TotalSpent
FROM
  (
    Customer_T AS C
    INNER JOIN Pet_T AS P ON C.CustomerID = P.CustomerID
  )
  INNER JOIN Appointment_T AS A ON P.PetID = A.PetID
WHERE
  A.Status = 'Completed'
GROUP BY
  C.CustomerID,
  C.FirstName,
  C.LastName
ORDER BY
  SUM(A.TotalCharge) DESC;
```

Query #2



Problem: List all the scheduled appointments with customer, pet and service

```
SELECT
  A.AppointmentID,
  A.AppointmentDateTime,
  C.FirstName,
  C.LastName,
  P.PetName,
  S.ServiceName,
  A.Status,
  A.TotalCharge
FROM
  (
    (
      Customer_T AS C
      INNER JOIN Pet_T AS P ON C.CustomerID = P.CustomerID
    )
    INNER JOIN Appointment_T AS A ON P.PetID = A.PetID
  )
  INNER JOIN Service_T AS S ON A.ServiceID = S.ServiceID
WHERE
  A.Status = "Scheduled"
ORDER BY
  A.AppointmentDateTime;
```

AppointmentID	AppointmentDateTime	FirstName	LastName	PetName	ServiceName	Status	TotalCharge
2	11/11/2024 2:00:00 PM	Dan	Ruiz	Rocky	Basic Bath	Scheduled	\$35.00
5	11/13/2024 3:00:00 PM	Reggie	Kaverga	Nala	Nail Trim	Scheduled	\$15.00
10	11/16/2024 11:00:00 AM	Maria	Lopez	Chiquita	Nail Trim	Scheduled	\$15.00
(New)							

Query #3

ServiceID	ServiceName	ServiceCategory	TimesBooked
2	Full Groom	Groom	5
1	Basic Bath	Bath	4
5	Flea Bath	Treatment	2
3	Nail Trim	Add-On	2
4	Teeth Cleaning	Add-On	1



Problem: Rank services by how many times they were booked (Most → to least)

Code:

```
SELECT
    S.ServiceID,
    S.ServiceName,
    S.ServiceCategory,
    COUNT(A.AppointmentID) AS TimesBooked
FROM
    Service_T AS S
    LEFT JOIN Appointment_T AS A ON S.ServiceID =
    A.ServiceID
GROUP BY
    S.ServiceID,
    S.ServiceName,
    S.ServiceCategory
ORDER BY
    Count(A.AppointmentID) DESC;
```

Query #4



Problem: Monthly Revenue summary (Total dollars per month)

Code:

```
SELECT
    YEAR(A.AppointmentDateTime) AS RevenueYear,
    MONTH(A.AppointmentDateTime) AS RevenueMonth,
    SUM(A.TotalCharge) AS MonthlyRevenue
FROM
    Appointment_T AS A
WHERE
    A.Status = 'Completed'
GROUP BY
    Year(A.AppointmentDateTime),
    Month(A.AppointmentDateTime)
ORDER BY
    YEAR(A.AppointmentDateTime),
    MONTH(A.AppointmentDateTime);
```

RevenueYear ▾	RevenueMonth ▾	MonthlyRevenue ▾
2024	11	\$465.00

Query #5



Problem: List Customers who have never booked an appointment yet

CustomerID ▾	FirstName ▾	LastName ▾	Phone ▾	Email ▾
3	Sophia	Kim	630-344-2345	sophia.kim@gn
*	(New)			

Code:

```
SELECT
  C.CustomerID,
  C.FirstName,
  C.LastName,
  C.Phone,
  C.Email
FROM
  (
    Customer_T AS C
    LEFT JOIN Pet_T AS P ON C.CustomerID =
P.CustomerID
  )
  LEFT JOIN Appointment_T AS A ON P.PetID = A.PetID
WHERE
  A.AppointmentID IS NULL;
```

Query #6



Problem: Average spending per customer (for completed appointments)

Code:

```
SELECT
    C.CustomerID,
    C.FirstName,
    C.LastName,
    SUM(A.TotalCharge) AS TotalSpent,
    COUNT(A.AppointmentID) AS NumAppointments,
    SUM(A.TotalCharge) / COUNT(A.AppointmentID) AS
AvgSpentPerAppt
FROM
    (Customer_T AS C
    INNER JOIN Pet_T AS P
        ON C.CustomerID = P.CustomerID)
    INNER JOIN Appointment_T AS A
        ON P.PetID = A.PetID
WHERE
    A.Status = 'Completed'
GROUP BY
    C.CustomerID, C.FirstName, C.LastName
ORDER BY
    SUM(A.TotalCharge) DESC;
```

CustomerID	FirstName	LastName	TotalSpent	NumAppoint	AvgSpentPer
10	Araceli	Garcia	\$130.00	2	65
9	Olivia	Martinez	\$95.00	2	47.5
5	Emily	Garcia	\$85.00	2	42.5
1	Maria	Lopez	\$70.00	1	70
8	Kevin	Johnson	\$35.00	1	35
4	Jason	Patel	\$30.00	1	30
7	Ana	Torres	\$20.00	1	20

Query #7



Problem: Services whose BasePrice is above the average BasePrice (subquery)

Code:

```
SELECT
    S.Service_ID,
    S.ServiceName,
    S.ServiceCategory,
    S.BasePrice
FROM
    Service_T AS S
WHERE
    S.BasePrice >
        (SELECT AVG(BasePrice) FROM Service_T)
ORDER BY
    S.BasePrice DESC;
```

ServiceID	ServiceName	ServiceCateg	BasePrice
2	Full Groom	Groom	\$65.00
5	Flea Bath	Treatment	\$40.00
(New)			

Query #8



Problem: Top 3 customer by number of appointment (using subquery)

Code:

CustomerID	FirstName	LastName	NumAppoint
1	Maria	Lopez	3
10	Araceli	Garcia	2
9	Olivia	Martinez	2
5	Emily	Garcia	2

```
SELECT TOP 3
    C.CustomerID,
    C.FirstName,
    C.LastName,
    COUNT(A.AppointmentID) AS NumAppointments
FROM
    (Customer_T AS C
     INNER JOIN Pet_T AS P
       ON C.CustomerID = P.CustomerID)
     INNER JOIN Appointment_T AS A
       ON P.PetID = A.PetID
GROUP BY
    C.CustomerID, C.FirstName, C.LastName
ORDER BY
    COUNT(A.AppointmentID) DESC;
```


Query #9



Problem: Average total charge per Service Category

Code:

```
SELECT
    S.ServiceCategory,
    Avg(A.TotalCharge) AS AvgChargePerAppt
FROM
    Service_T AS S
    INNER JOIN Appointment_T AS A ON S.ServiceID =
    A.ServiceID
WHERE
    A.Status = 'Completed'
GROUP BY
    S.ServiceCategory
ORDER BY
    AVG(A.TotalCharge) DESC;
```

ServiceCateg ▼	AvgChargePe ▼
Groom	\$68.00
Add-On	\$30.00
Bath	\$27.50
Treatment	\$20.00

Query #10

Code:

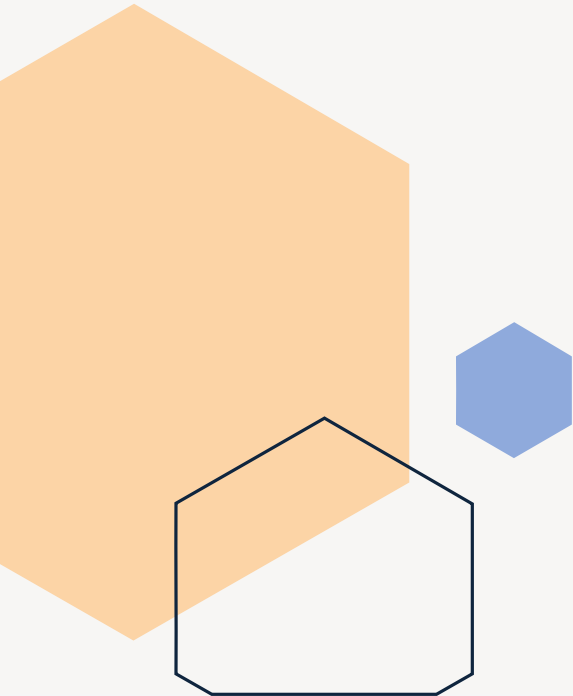
CustomerID	FirstName	LastName
1	Maria	Lopez
5	Emily	Garcia
8	Kevin	Johnson
9	Olivia	Martinez
10	Araceli	Garcia



Problem: UNION- Customer who are “VIP” how have either spent more than \$30 and visited at least twice

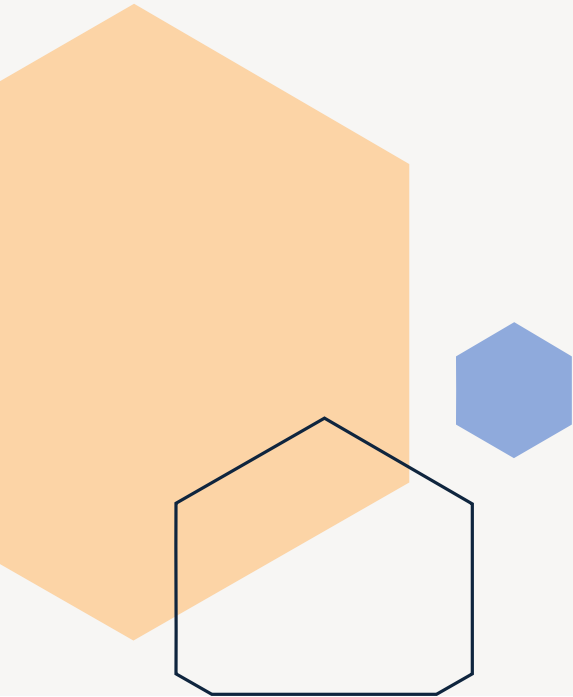
```
SELECT
    C.CustomerID,
    C.FirstName,
    C.LastName
FROM
    (
        Customer_T AS C
        INNER JOIN Pet_T AS P ON C.CustomerID = P.CustomerID
    )
    INNER JOIN Appointment_T AS A ON P.PetID = A.PetID
WHERE
    A.Status = 'Completed'
GROUP BY
    C.CustomerID,
    C.FirstName,
    C.LastName
HAVING
    SUM(A.TotalCharge) > 30
UNION
SELECT
    C2.CustomerID,
    C2.FirstName,
    C2.LastName
FROM
    (
        Customer_T AS C2
        INNER JOIN Pet_T AS P2 ON C2.CustomerID = P2.CustomerID
    )
    INNER JOIN Appointment_T AS A2 ON P2.PetID = A2.PetID
GROUP BY
    C2.CustomerID,
    C2.FirstName,
    C2.LastName
HAVING
    COUNT(A2.AppointmentID) > 2;
```

Business Insights



- Q1: Figuring out what customer spent the most and by how much can aid a business in knowing their target audience. We can also send out special promotions to the customer to establish a stronger customer relationship.
- Q2: Listing all upcoming scheduled appointments help businesses with scheduling enough staff. Also helps in buying enough inventory to ensure sufficient supplies are in stock.
- Q3: Ranking the services by how many were booked helps a business know which service is popular. We can expand on the popular service. Alternatively we get information on the unpopular service and can slash it if it is not bringing in sufficient revenue.
- Q4: The monthly revenue query helps a business know how it's doing financially. If we are working in a deficit we can make the proper adjustments to increase revenues.
- Q5: Listing customers who have never booked an appointment can help us send out promotions to the customers we have on file to get them to do business with us.

Business Insights



- Q6: Figuring out the average spending price per customer can help figure out how our prices impact our revenues.
- Q7: Figuring out what base price is above average helps in seeing how the prices are spread out.
- Q8: The top 3 customers represent our most frequent and engaged clients, which suggests they have a strong ongoing need for grooming services and high loyalty to the salon.
- Q9: Highlighting the relative profitability of different service categories can help identify what services generate more revenue per visit.
- Q10: Identifying our VIP customers helps give valuable insight into our business. Analyzing out high spenders behavior and high repeat usage. We could provide perks to out most loyal customers like free nail trims after 3 visits. Help increase our revenues.